

Course Learning Outcomes: Course Outline:

- Coulomb's Law,
- Superposition of forces
- conservation of charge/charge quantization,
- Electric field & Superposition of Electric fields,
- Ring of charge,
- Disk of charge,
- Dipole in an electric field.

Chapter 21, University Physics by
Young and Freedman

- The flux of vector field,
- The flux of electric field,
- Gauss Law and its general form

Chapter 22, University Physics by
Young and Freedman

- Electric potential energy
- Electric potential energy of two point charges,
- Electric potential energy with several point charges,
- Electric potentials
- Potential due to point & continuous charge distribution
- Calculating the potential from the field,
- Equipotential surfaces,
- Calculating the field from the potential.

Chapter 23, University Physics by
Young and Freedman

- Electric current,
- Current density,
- Resistance,
- Resistivity and conductivity,
- Superconductivity
- Ohm's law and its forms

Chapter 25, University Physics by
Young and Freedman

- Magnetic field lines & magnetic flux
- The Hall effect

Chapter 27, University Physics by
Young and Freedman

Lab

- KCL and KVL
- Diode and biasing
- V-I characteristics of a diode
- Half-wave Rectifiers
- Full-wave Rectifiers

Chapter 1, Electronic Devices by
Floyd

Chapter 2, Electronic Devices by
Floyd

Note: All the topics discussed during the lectures are included in the syllabus.